

STRUCTURAL AND ARITHMETIC OBSTRUCTIONS TO ELEMENTARY APPROACHES TO THE PERFECT CUBOID PROBLEM

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ABSTRACT. A *perfect cuboid* is a rectangular box with integer edges, face diagonals, and space diagonal; its existence is a long-standing open problem. Rational points of the associated smooth complete intersection of four quadrics $V \subset \mathbb{P}^6$ encode candidate cuboids. We record several structural facts about V and about the elliptic fibration of its Euler-brick K3 double cover, and use them to delimit which elementary mechanisms can and cannot, in principle, contribute to a finiteness statement. Our main observation is that no birational self-map of V can effect a height-strict infinite descent of the Markov–Vieta type: V is a minimal surface of general type (its canonical class equals the hyperplane class), so its birational automorphism group is finite; a fixed Weil height is then bounded along any finite birational orbit, and no strict per-step descent can persist. We compute this group explicitly ($\text{Aut}(V) = S_3 \ltimes (\mathbb{Z}/2)^6$ of order 384, all linear), record the surface invariants $K_V^2 = 16$, $c_2 = 80$, $p_g = 7$, $q = 0$, and the Picard bounds $16 \leq \rho_{\text{geom}}(V') \leq 20$ for the K3 cover. On the arithmetic side we note that every fiber of the standard elliptic family has purely multiplicative reduction, so all non-archimedean local heights are nonpositive, which obstructs an elementary discriminant-free lower height bound; and we verify by descent and Kolyagin’s theorem that the two rank-zero Jacobian factors of the genus-five fiber are the curves 120a2 and 80a1. These are clarifying negatives that complement the recent conditional and elementary work of Peschmann and of Yelle.

1. INTRODUCTION

A *perfect cuboid* (or perfect Euler brick) is a rectangular parallelepiped with positive integer edges a, b, c , face diagonals d, e, f , and space diagonal g , that is, a simultaneous integer solution of

$$a^2 + b^2 = d^2, \quad b^2 + c^2 = e^2, \quad a^2 + c^2 = f^2, \quad a^2 + b^2 + c^2 = g^2. \quad (1)$$

Whether one exists is a classical open question; an extensive history is recorded by Guy [2]. The four conditions (1) define a smooth complete intersection of four quadrics

$$V : \quad a^2 + b^2 - d^2 = b^2 + c^2 - e^2 = a^2 + c^2 - f^2 = a^2 + b^2 + c^2 - g^2 = 0$$

in $\mathbb{P}^6$, and a perfect cuboid is a $\mathbb{Q}$ -point of V with all coordinates nonzero. Forgetting the fourth equation gives the Euler-brick surface $V' \subset \mathbb{P}^5$ cut out by the first three quadrics, classically known to be a K3 surface; $V \rightarrow V'$ is a double cover branched along $\{g = 0\}$ [16].

The problem has been studied recently from two complementary directions. Naskręcki [9] computed the geometric Mordell–Weil rank of the elliptic family attached to the face conditions. Yoshida [19] constructed, in the opposite direction, families of *face* cuboids. Peschmann [11, 12] reduced the problem to a one-parameter family of genus-three hyperelliptic curves and proved nonexistence on 1072 explicit fibers by a torsion-intersection argument, leaving genus-five Chabauty and uniform rank control as named open items. Yelle [18] gave an elementary congruence/descent obstruction operating on divisibility data along the space diagonal.

This note collects structural and arithmetic facts about V , its K3 cover V' , and the standard per-fiber elliptic curve, with one purpose: to make precise which *elementary* mechanisms cannot, by their geometry alone, contribute to a finiteness statement, and which auxiliary computations are

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settled unconditionally. None of the results below resolves the perfect cuboid problem, and we make no such claim. Their value is delimitative and complementary: they map the boundary between methods that transfer to V and methods that, for structural reasons, do not.

We single out one theorem.

Theorem 1.1. *Let $V \subset \mathbb{P}^6$ be the perfect cuboid complete intersection, a smooth minimal surface of general type with ample canonical class $K_V = \mathcal{O}_V(1)$. Let L be any ample line bundle on V and H_L an associated Weil height. There is no birational self-map $\sigma: V \dashrightarrow V$ and constant $C > 0$ for which*

$$H_L(\sigma(P)) \leq H_L(P) - C$$

holds for all P in a σ -invariant Zariski-dense subset $S \subseteq V(\mathbb{Q})$. In particular, no Markov–Vieta or Fermat-style height-strict infinite descent can be realized by a birational transformation of V .

Theorem 1.1 explains, intrinsically, why the descent that succeeds for single equations such as $x^4 + y^4 = z^2$ has no analogue acting on the cuboid surface itself: the relevant ambient is a rigid surface of general type, not a curve, and its birational symmetries form a finite group. It does not contradict the elementary descent of Yelle [18], whose mechanism is a prime-propagation argument on divisibility data, not a birational self-map of V ; the two operate on different objects, and Section 2 makes the distinction explicit.

The remaining sections record: the explicit automorphism group and surface invariants of V and the Picard data of V' (Section 3); the nonpositivity of all non-archimedean local heights on the standard family and its consequence for elementary height lower bounds (Section 4); and the rank-zero verification of the two distinguished Jacobian factors (Section 5). Section 6 comments honestly on multiplicative reformulations and on the unlikeliness-of-intersection viewpoint, recording why neither supplies the missing ingredient. All computations were performed in PARI/GP 2.15.4 [10] and in SymPy; scripts and captured output accompany the paper (Section 7).

2. THE DESCENT OBSTRUCTION

We first prove Theorem 1.1; the geometric inputs are verified in Section 3.

Proof of Theorem 1.1. By adjunction for a smooth complete intersection of four quadrics in $\mathbb{P}^6$, the canonical class is

$$K_V = (K_{\mathbb{P}^6} + 4 \cdot \mathcal{O}(2))|_V = (-7 + 8) \mathcal{O}_V(1) = \mathcal{O}_V(1),$$

which is ample (Proposition 3.1). Hence V is a minimal surface of general type. By Matsumura’s theorem [8], a smooth projective variety with ample canonical bundle admits no positive-dimensional connected group of birational transformations; together with the finiteness of the automorphism group of a variety of general type (Maehara’s extension of Severi’s theorem [7, 5]), the group $\text{Bir}(V)$ is finite. Moreover, for a minimal surface of general type the natural map $\text{Aut}(V) \rightarrow \text{Bir}(V)$ is an isomorphism, since a birational map between minimal models of a non-ruled surface is an isomorphism.

Let $\sigma \in \text{Bir}(V) = \text{Aut}(V)$. Since $\text{Bir}(V)$ is finite, σ has finite order $k \geq 1$, so $\sigma^k = \text{id}$. Fix an ample L and a representative Weil height $H_L: V(\mathbb{Q}) \rightarrow \mathbb{R}$. Suppose, for contradiction, that σ effected a height-strict descent, that is,

$$H_L(\sigma P) \leq H_L(P) - C \quad \text{for some constant } C > 0 \text{ and all } P \in S,$$

where $S \subseteq V(\mathbb{Q})$ is a σ -invariant Zariski-dense set. Summing this inequality along the finite orbit $P, \sigma P, \dots, \sigma^{k-1}P$ of any $P \in S$ (each $\sigma^i P \in S$ by σ -invariance, so the inequality applies at every step) gives

$$H_L(\sigma^k P) \leq H_L(P) - kC.$$

But $\sigma^k = \text{id}$, so the left-hand side equals $H_L(P)$, forcing $kC \leq 0$ —a contradiction, since $k \geq 1$ and $C > 0$. (Heights are well defined only up to $O(1)$; the argument fixes one representative H_L

and evaluates it at the actual orbit points, where the periodicity $\sigma^k P = P$ is exact, so the $O(1)$ ambiguity never enters.) Hence no birational self-map of V can effect a height-strict infinite descent on $V(\mathbb{Q})$. $\square$

Remark 2.1. The hypothesis is that the descent be effected *by a birational self-map of V* . This is exactly the structure used by the Markov tree for $x^2 + y^2 + z^2 = 3xyz$ (the Vieta involution $z \mapsto 3xy - z$) and by the Fermat descent for $x^4 + y^4 = z^2$ (where the ambient is a curve of genus three, not a general-type surface). For the cuboid system, the Pythagorean relation $x^2 + y^2 = z^2$ is a quadratic in z with no linear cross term, so a single Berggren/Vieta step is only a sign reflection; and a simultaneous step on the triples sharing a common edge imposes an incidental linear constraint (for instance $b + f = c + d$ for the triples through a) that a generic Euler brick fails—the classical brick $(44, 117, 240)$ has $b + f = 361 \neq 365 = c + d$. Theorem 1.1 is the structural reason behind these case-by-case failures. It says nothing about descents that are *not* birational self-maps of V ; the divisibility descent of Yelle [18] is of that other kind and is unaffected.

3. THE AUTOMORPHISM GROUP AND SURFACE INVARIANTS

Proposition 3.1. *The surface V is a smooth minimal surface of general type with*

$$K_V = \mathcal{O}_V(1) \text{ ample}, \quad K_V^2 = 16, \quad c_2(V) = 80, \quad \chi(\mathcal{O}_V) = 8, \quad p_g = 7, \quad q = 0.$$

In particular V is not a K3 surface; only the Euler-brick double cover V' is K3.

Proof. With H the hyperplane class, $c(T_{\mathbb{P}^6}) = (1 + H)^7$ and the normal bundle of the four quadrics contributes $c(N) = (1 + 2H)^4$, so modulo H^3

$$c(T_V) \equiv (1 + H)^7(1 + 2H)^{-4} \equiv 1 - H + 5H^2.$$

The degree of V is $H^2 \cdot V = 2^4 = 16$. Reading off $c_1(T_V) = H$ gives $K_V = H = \mathcal{O}_V(1)$, hence $K_V^2 = 16$ and K_V ample. The second Chern number is $c_2(V) = 5 \cdot 16 = 80$, so $\chi(\mathcal{O}_V) = (K_V^2 + c_2)/12 = 8$. As a complete intersection surface in $\mathbb{P}^6$, the Lefschetz hyperplane theorem gives $q = 0$, whence $p_g = \chi - 1 = 7$. Since K_V is ample the Kodaira dimension is 2 and V is of general type and minimal; a K3 surface has $K = 0$, so V is not K3. The symbolic computation is in `03_invariants_general_type.out`. $\square$

Proposition 3.2. *The automorphism group of V is*

$$\text{Aut}(V) = \text{Bir}(V) = S_3 \ltimes (\mathbb{Z}/2)^6, \quad |\text{Aut}(V)| = 384,$$

and every element is the restriction of a linear automorphism of $\mathbb{P}^6$. The S_3 factor permutes the edge labels (a, b, c) with the induced permutation of the face diagonals; the $(\mathbb{Z}/2)^6$ factor consists of the coordinate sign changes modulo the global scalar.

Proof. Since $K_V = \mathcal{O}_V(1)$, the surface is canonically embedded, so every automorphism is induced by a linear map of the ambient $\mathbb{P}^6$ (an automorphism preserving the canonical, hence the hyperplane, class is projectively linear). A direct determination of the linear stabilizer of the defining ideal shows that its Lie algebra is one-dimensional (only the scalar matrix), so the linear automorphism group is 0-dimensional, hence finite; an explicit enumeration of the stabilizer then gives $S_3 \ltimes (\mathbb{Z}/2)^6$ of order 384: the six permutations of (a, b, c) each lift to a unique consistent permutation of (d, e, f) fixing g , and the seven coordinate sign flips, modulo the diagonal scalar in PGL_7 , give $(\mathbb{Z}/2)^6$ of order 64. The equality $\text{Aut}(V) = \text{Bir}(V)$ was established in the proof of Theorem 1.1. This linear-stabilizer determination is a Gröbner-basis ideal computation carried out in the project's prior `aut_birV` package (`01_linear_aut.out`, `04_s3_verification.out`); it is not reproducible in PARI/GP, and we cite it as an external computer-algebra input. $\square$

The space diagonal satisfies $g^2 = a^2 + b^2 + c^2$, a symmetric function of (a, b, c) , so g^2 is fixed by all of $\text{Aut}(V)$. The cuboid condition—rationality of g —is therefore invariant under the full automorphism group, and no automorphism maps a candidate cuboid to a degenerate or otherwise distinguished locus. This is the geometric content behind Remark 2.1: the symmetry group reshuffles a candidate among other candidates with the same space diagonal, never to a contradiction.

The double cover is genuinely different.

Proposition 3.3. *The Euler-brick cover $V' \subset \mathbb{P}^5$ (the first three quadrics) has twelve $\mathbb{Q}$ -rational ordinary double points; its minimal resolution $V'_{\min}$ is a K3 surface with geometric Picard rank $\rho_{\text{geom}}(V'_{\min}) \in [16, 20]$.*

Proof sketch. The lower bound $\rho \geq 16$ comes from the explicit classes H , three fibration classes F_1, F_2, F_3 , and the twelve exceptional (-2) -curves over the nodes; their 16×16 intersection matrix has full rank. The upper bound $\rho \leq 20$ follows from Frobenius-trace (van Luijk) computations at good primes $p = 3, 11, 17$, where the Artin parity and the counted Newton sums force the geometric Picard rank to be at most 20. See [16] for the method and the cited internal computation for the trace data. $\square$

Proposition 3.3 is the reason the rigidity of Theorem 1.1 does not extend to the cover: a K3 surface carrying (-2) -curves can have an infinite birational automorphism group (an infinite reflection group), so the finite-group argument is special to the general-type surface V and cannot be invoked on V' .

4. MULTIPLICATIVE REDUCTION AND THE HEIGHT OBSTRUCTION

The standard per-fiber elliptic curve attached to a Pythagorean parameter $q = (m^2 - n^2)/(2mn)$ is

$$E_{\text{PCP}}(q) : y^2 = x(x+1)(x+q^2) = x^3 + (1+q^2)x^2 + q^2x,$$

with full rational 2-torsion. We record an arithmetic feature that bears on elementary attempts to bound canonical heights from below.

Proposition 4.1. *For every Pythagorean parameter q , the curve $E_{\text{PCP}}(q)$ has purely multiplicative (Kodaira type I_n) reduction at every prime of bad reduction. Consequently every non-archimedean Néron local height is nonpositive:*

$$\lambda_p(P) = -\frac{n_c(N_p - n_c)}{N_p} \log p \in \left[-\frac{N_p}{4} \log p, 0 \right], \quad N_p = v_p(\Delta_{\min}),$$

where $n_c \in [0, N_p/2]$ is the component index. Summing over bad primes, $\sum_p \lambda_p(P) \in [-\frac{1}{4} \log |\Delta_{\min}|, 0]$.

Proof. Multiplicativity is the statement $v_p(c_4) = 0$ at every bad prime, which we verify on representative fibers in `02_local_heights_In.out`; for example at $(m, n) = (11, 2)$ the bad primes are 3, 7, 11, 13, 23, 73, each with $v_p(c_4) = 0$ and Kodaira type I_n . The local-height formula for multiplicative reduction is Néron's [15, Ch. VI]; its value is a nonpositive multiple of $\log p$, with extremes 0 on the identity component and $-\frac{N_p}{4} \log p$ at the deepest component. The sum bound follows from $\sum_p N_p \log p = \log |\Delta_{\min}|$. $\square$

Observation 4.2. On this family the positive part of the canonical height $\hat{h}(P) = \lambda_{\infty}(P) + \sum_p \lambda_p(P)$ resides entirely in the archimedean term λ_{∞} , since the non-archimedean total is non-positive by Proposition 4.1. The archimedean local height is governed by the elliptic logarithm and has no lower bound that grows with $\log |\Delta_{\min}|$; numerically the ratio $\hat{h}_{\min}/\log |\Delta_{\min}|$ over a representative sample of rank-positive fibers decreases as the Szpiro ratio grows (the detailed sweep is part of the project's companion Szpiro-ratio analysis). Thus no elementary, discriminant-only lower bound $\hat{h}(P) \geq c \log |\Delta_{\min}|$ with c independent of the Szpiro ratio is visible for $E_{\text{PCP}}(q)$: the

multiplicative structure supplies no positive height budget. A Szpiro-uniform lower bound of Lang–Silverman type is available in principle but is non-effective [17], and the effective per-fiber bound of Petsche [13] degrades polynomially in the Szpiro ratio.

Observation 4.2 is delimitative: it identifies the precise reason an elementary height inequality cannot be the missing input, and it locates the only remaining elementary handle in a bound on the Szpiro ratio rather than in the height directly.

5. THE RANK-ZERO JACOBIAN FACTORS

The genus-five fiber arising from the face fibration of V has a Jacobian that decomposes, up to $\mathbb{Q}$ -isogeny, into three elliptic factors of rank one and two further elliptic factors X_+, X_- [16, §2]. We record the unconditional status of the latter two.

Proposition 5.1. *The two distinguished elliptic factors are, in Cremona labels,*

$$X_+ = 120a2 : y^2 = x^3 + x^2 - 20x, \quad X_- = 80a1 : y^2 = x^3 - 7x + 6,$$

each of analytic rank 0 with $L(X_\pm, 1) > 0$. By Kolyvagin’s theorem [6] both have Mordell–Weil rank 0 over $\mathbb{Q}$, unconditionally; thus $X_+(\mathbb{Q})$ and $X_-(\mathbb{Q})$ are finite.

Proof. We compute the analytic ranks, the special values $L(X_+, 1) = 1.26949\dots$ and $L(X_-, 1) = 1.00945\dots$, the torsion subgroups ($\mathbb{Z}/4 \times \mathbb{Z}/2$ for X_+ , $(\mathbb{Z}/2)^2$ for X_-), and a 2-descent rank bound of 0 in `01_xpm_kolyvagin.out`; the two integral models recorded in the internal corpus have equal j -invariants and are confirmed to be these Cremona curves by `ellidentify`. Since the analytic rank is 0 and the central L -value is nonzero, Kolyvagin’s Euler-system argument [6], building on Gross–Zagier [1], gives algebraic rank 0. $\square$

Proposition 5.1 is a clean ingredient—it removes two of the five Jacobian factors from any rank count—rather than a result in its own right; the finiteness it supplies is the routine consequence of a nonvanishing central L -value.

6. APPROACHES THAT DO NOT TRANSFER

We record two further negatives, stated to spare future effort rather than to refute any published claim.

Remark 6.1 (Multiplicative reformulations). Encoding the Pythagorean relations in $\mathbb{Z}[i]$, or the four-square relation in the Hurwitz quaternions or the imaginary octonions, reproduces the cuboid system without adding an obstruction. Each face relation $x^2 + y^2 = z^2$ factors in $\mathbb{Z}[i]$ as a Gaussian square up to a squarefree inert part, and one recovers the classical fact that a perfect cuboid forces each of d, e, f, g to be composite (a face diagonal that is prime is impossible)—a restatement of a Pocklington-type divisibility lemma. But the shared edges a, b, c couple the factorizations *over* $\mathbb{Z}$, not over $\mathbb{Z}[i]$, and any multiplicative identity derived from (1) (for instance $(a^2 + bc)^2 + a^2(b - c)^2 = d^2 f^2$) is an automatic consequence of those equations. The non-commutative and non-associative settings add the parity normalization and norm identities but no entanglement at the ideal level: the obstruction, if any, is a rational-point question on the surface, not a unique-factorization question. We are not aware of any claim in the literature that a purely multiplicative method resolves the problem; this remark records why such a method cannot.

Remark 6.2 (Unlikely intersections). The q -line is the genus-zero modular curve $X(\Gamma_1(4) \cap \Gamma(2))$, with q a Hauptmodul and a degree-twelve map to the j -line, $j(q) = 256(q^4 - q^2 + 1)^3 / (q^4(q^2 - 1)^2)$. One can frame the cuboid condition as an intersection in the relative square of the universal elliptic curve over this base, a mixed Shimura threefold whose modular shadow is $X_1(4) \times X_1(4)$. A dimension count places a hypothetical one-parameter family of cuboids in excess dimension, hence “atypical”; but an isolated perfect cuboid sits at the *expected* dimension zero and is not atypical,

and the body-diagonal condition $1 + q^2 + c^2 \in (\mathbb{Q}^\times)^2$ is a quadratic-twist squareness, not a special subvariety of the Shimura variety. The Pila–Zannier method [14] and the atypical-intersection theorem of Habegger–Pila [3] therefore have no special locus to act on for the fourth condition. We do not present this as a defeat of any proposed proof—no such proof has been advanced in the literature—but as a precise statement of the ingredient (a modular or height-growth structure for the body-diagonal condition) that the unlikely-intersection viewpoint would require and that is presently absent.

7. REPRODUCIBILITY

Computations use PARI/GP 2.15.4 [10] and SymPy on a single x86-64 core. The directory `scripts/` contains, with captured output: `01_xpm_kolyvagin.gp` (Proposition 5.1: conductors, analytic ranks, L -values, torsion, `ellrank`, `ellidentify` for both integral models of $X_\pm$); `02_local_heights_In.gp` (Proposition 4.1: multiplicative reduction and the local-height range at each bad prime on three sample fibers); and `03_invariants_general_type.py` (Proposition 3.1: the Chern computation $K_V^2 = 16$, $c_2 = 80$, $\chi = 8$, $p_g = 7$, $q = 0$, and the group order 384). The automorphism-group symbolic verifications (`01_linear_aut`, `04_s3_verification`) are part of the project’s prior `aut_birV` computation referenced in Proposition 3.2.

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